

Generation Interconnection System Impact Study Report

For Queue Project Q21-1665

Bel Air 138 kV

June 01, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1665. The Interconnection Customer proposes a 51.0 MW solar photovoltaic facility interconnecting at the Bel Air 138 kV substation in the North Carolina area. The request was received on February 10, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 531 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$184,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Chipola / Spring St 138.0 kV
Contingency: Mitchell Dam / Skaggs Island Jct 2 345.0 kV
Loading: 108.1% (Rating: 234 MVA)
DFAX: 7.2%
Pre-project loading: 106.5%
2. Facility: Natchez / Chancellor Row 138.0 kV L
Contingency: Mitchell Dam / Skaggs Island Junction 2 345.0 kV
Loading: 100.9% (Rating: 508 MVA)
DFAX: 6.3%
Pre-project loading: 100.3%
3. Facility: Natchez / Lakeside 138.0 kV
Contingency: Mitchell Dam / Skaggs Island Jct 2 345.0 kV L
Loading: 100.6% (Rating: 531 MVA)
DFAX: 6.3%
Pre-project loading: 99.9%
4. Facility: Hull St Road / Harvard 138.0 kV
Contingency: Mitchell Dam / Skaggs Island Junction 2 345.0 kV
Loading: 100.1% (Rating: 574 MVA)
DFAX: 9.6%
Pre-project loading: 99.3%
5. Facility: Nauvoo Substation / Birchwood 345.0 kV L
Contingency: Fredericksburg / Dunn Siding 345.0 kV
Loading: 100.1% (Rating: 2,444 MVA)
DFAX: 15.7%
Pre-project loading: 99.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chipola / Spring St 138.0 kV	Reconductor line	\$2,722,000	1% (\$28,000)
Chancellor Row / Natchez 138.0 kV L	Reconductor line	\$1,399,000	1% (\$8,000)
Lakeside / Natchez 138.0 kV	Reconductor line	\$1,126,000	1% (\$7,000)
Hull St Road / Harvard 138.0 kV	Reconductor line	\$548,000	1% (\$5,000)
Nauvoo / Birchwod 345.0 kV	Reconductor line	\$23,964,000	1% (\$136,000)
	Total allocated to project		\$184,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.